

Ans: B

Q9

At X:

$$N + mg = mv^2/r$$

$$N = mv^2/r - mg$$

$$v = \sqrt{rg} \text{ when } N \approx 0$$

Since GPE at X and Y is the same,

KE at X = KE at Y.

At Y:

$$mg - N = mv^2/2r$$

$$N = mg - m(rg)/2r$$

$$N = mg/2$$

Ans: B

Q10